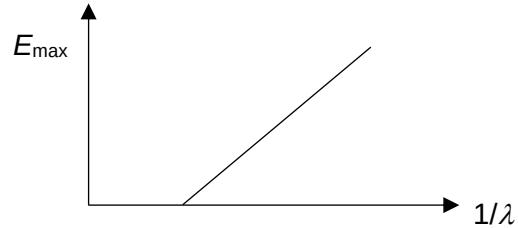


Q5(a)

The threshold frequency is the minimum frequency of incident electromagnetic radiation required to cause electrons to be emitted from a metal.

Q5(b)(i)



Q5(b)(ii)

gradient of line = hc

where h is Planck's constant and c is the speed of light.

Q5(b)(iii)

$E_{\max}$ is for surface photoelectrons, where the electrons require the least energy to be emitted.

Electrons below the surface require more energy from the incident radiation to bring them to the surface for emission. This means there is less energy from incident photon energy transformed into KE of the escaping photoelectrons.

[Accept electrons in the deeper layers of the metal will encounter collisions with intervening atoms on the way to the metal surface, resulting in the photoelectrons having less KE.]

Q5(b)(iv)

$$\text{Intensity } I = \frac{N \frac{hc}{\lambda}}{tA} = \left(\frac{N}{t} \right) \frac{hc}{\lambda A} \dots\dots(1)$$

where $\frac{N}{t}$ is the rate of incident photons.

For intensity to be kept constant, $\frac{N}{t}$ is reduced when λ is reduced.

This reduces the photocurrent since there are fewer photons incident on the metal per unit time.

Q6(a)(i)